

Joshua D. Arnow

SOFTWARE ENGINEER

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CAREER OBJECTIVE

Highly motivated full-stack software engineer and AI/ML researcher. Seeking to make a meaningful impact with a passionate team.

PROFESSIONAL EXPERIENCE

Founder & CEO

12/2025 – Present

Lumavant, LLC

- Create a novel, full-stack, and multi-platform application offering intuitive recipe creation, user experience, and sharing.
 - Leverage Apple's Foundation Models to develop a powerful AI assistant serving home cooks and culinary enthusiasts.
 - Design and implement a seamless user experience across platforms using Swift, Kotlin, React, Go, and PostgreSQL.
 - Integrate millions of recipes via extensive web scraping to solve the cold start problem and provide instant value to users.
- Build responsive e-commerce platforms in React for clients determined to expand their digital presence and drive online sales.

Team Lead

5/2023 – 12/2025

Software Engineer

6/2022 – 5/2023

Two Fish Creative, Inc.

- Led cross-functional development teams, managed the full software development life cycle (SDLC), and collaborated with product managers and UX/UI designers to deliver complex, high-quality client projects.
 - Implemented Agile/Scrum processes, performed rigorous code reviews, and managed timelines for timely feature delivery.
 - Architected and built full-stack platforms for clients using TypeScript, React, Python/Flask, MySQL/MariaDB, and AWS.
 - Created and independently published native iOS and Android applications and designed server/database integrations.
 - Migrated monolithic legacy platforms to modern service-oriented architectures (microservices) utilizing Docker/Kubernetes.
 - Ensured high-quality, scalable software components by managing CI/CD pipelines and writing unit and integration tests.
- Engineered and delivered custom software solutions and architectural upgrades across diverse client industries such as logistics, e-commerce, surrogacy, and nightlife entertainment.
- Regularly exceeded client expectations, securing contract renewals at higher rates often by revitalizing stalled projects and delivering critical features that previous teams could not.

TECHNICAL SKILLS

Programming Languages

JavaScript, TypeScript, Python, Java, C/C++, Rust, Swift, Go, HTML5, SCSS/CSS3, SQL

AI & Machine Learning

PyTorch, TensorFlow, Pandas, Scikit-Learn, NumPy, Exploratory Data Analysis, Data Visualization, Neural Networks (CNNs, RNNs, Transformers, GANs), Computer Vision, Natural Language Processing, Foundation Models, Data Augmentation, Transfer Learning

Frameworks & Libraries

React.js, React Native, Next.js, Express.js, Flask, Tailwind CSS, ChartJS, Firebase, Mapbox GL JS, Pytest, Three.js, Framer Motion

Databases/ORMs

PostgreSQL, MySQL, MongoDB, MariaDB, SQLAlchemy, SQLx, GORM

Cloud Computing, Containerization, & CI/CD

AWS (EB, EC2, S3, RDS, EKS, Lambda), Docker, Kubernetes, Slurm, High-Performance Computing (HPC), GitHub Actions, Cloudflare

Miscellaneous

Scrum/Agile Methodologies, Figma, Business Process Modeling, Technical Strategy, System Architecture, Camunda

EDUCATION

Pace University

New York, NY

Master of Science in Computer Science, summa cum laude — Concentration: Artificial Intelligence

GPA: 4.0/4.0 — July 2026

Tulane University

New Orleans, LA

Bachelor of Science in Economics, cum laude

GPA: 3.7/4.0 — May 2016

Bachelor of Arts in Political Economy, cum laude

GPA: 3.7/4.0 — May 2016

AWARDS

Program Award — M.S. in Computer Science

Pace University — May 2026

- Recognized for outstanding academic achievement. Awarded to the top-performing student in the graduating class.

RESEARCH & PUBLICATIONS

OmniFusion: A Hybrid Deep Learning Foundation Model for Skin Cancer Diagnostics

Master's Thesis

Pace University
6/2025 – 5/2026

- Engineered a state-of-the-art computer vision model achieving 95.01% AUROC by dynamically fusing Vision Transformers with Convolutional Neural Networks (CNNs) via PyTorch.
 - Designed and executed a robust distributed training pipeline across Slurm HPC clusters, processing massive, extremely imbalanced dermatological datasets exceeding 400,000 images. Utilized a strict 10-fold cross-validation strategy to rigorously validate model generalizability and isolate the performance impacts of training set multimodality.
 - Developed a comprehensive medical image preprocessing pipeline, standardizing input sizes and applying Contrast Limited Adaptive Histogram Equalization (CLAHE), the DullRazor algorithm for hair artifact removal, and bilateral filtering to reduce noise while preserving critical lesion borders.
 - Implemented an in-place, randomized spatial and pixel-level augmentation pipeline alongside a linear probing then fine-tuning (LP-FT) strategy, matching state-of-the-art performance while drastically reducing training time and storage footprint. Incorporated complex cropping, geometric transformations, color jittering, and Gaussian noise.
 - Accomplished numerous research contributions, including: (1) proving the efficacy of a novel hybrid CNN/Transformer classifier architecture via statistical analysis, (2) empirically validating the performance impact of multimodal training data, and (3) demonstrating the cost-effectiveness of the model using Incremental Cost-Effectiveness Ratio (ICER) analysis.
 - Benchmarked the model's performance against existing state-of-the-art dermatology models, preparing the methodology and empirical findings for submission to top-tier medical and digital health journals.

SELECTED PROJECTS

Counterwait (Dining Order Processing System)

Tech Stack: *Rust, Axum, Leptos, PostgreSQL, SQLx, Docker*

Lumavant, LLC
5/2026 – Present

- Architecting and developing a full-stack dining order processing system to facilitate at-table mobile customer ordering.
 - Engineering a high-concurrency, memory-safe backend API using Rust and Axum to ensure scalability and reliability.
 - Developing a reactive, WebAssembly-based client interface using the Leptos framework, enabling seamless, low-latency at-table customer ordering.
 - Architecting a real-time data synchronization pipeline utilizing Server-Sent Events (SSE) to seamlessly integrate multi-POS systems, ensuring accurate, up-to-the-second menu, inventory, and order state management.
 - Designing and optimizing asynchronous database operations using PostgreSQL and SQLx, ensuring robust transaction handling and data integrity for high-volume hospitality environments.

Good Greek CRM

Tech Stack: *TypeScript, React Native, Python, Flask, MySQL, AWS (EC2, S3, EKS), Docker, Kubernetes*

Two Fish Creative, Inc.
9/2024 – 10/2025

- Architected and developed a custom enterprise logistics CRM to replace an inefficient legacy system, servicing an organization generating over \$100M in annual revenue. The new platform is projected to generate \$4.8 million in new revenue from reclaimed sales time and save \$672,000 in annual operational costs.
 - Navigated dynamic project scopes and worked through major architectural pivots with the development team, ensuring the successful delivery of proprietary features despite shifting client requirements.
 - Engineered a high-performance backend API using Flask, seamlessly integrating with MySQL to handle complex, high-volume operational data across multiple franchise locations.
 - Developed dynamic and responsive frontends utilizing React Native and Jinja, optimizing user workflows for sales, dispatching, accounting, and quality assurance teams.
 - Designed an intelligent job estimation engine featuring a comprehensive custom inventory module, dynamically calculating cubic footage, weight, moving duration, and resource requirements (trucks and crew).
 - Built a sophisticated dispatching and resource allocation system featuring an interactive visual calendar, enabling real-time drag-and-drop scheduling, multi-stop long-distance routing, and conflict resolution.
 - Automated critical business logic, including a robust sales commission engine and complex post-move quality assurance workflows, significantly reducing manual administrative overhead.
 - Established scalable CI/CD pipelines via GitHub Actions and managed AWS infrastructure to ensure high availability and reliable deployment cadences.

Multi-Class Natural Language Processing (NLP) News Classifier

Tech Stack: *Python, Pandas, Scikit-Learn, NumPy, Matplotlib, NLTK/spaCy*

Pace University
4/2026 – 5/2026

- Engineered an end-to-end multi-class text classification pipeline to categorize a dataset of over 81,000 HuffPost news articles into distinct topical domains.
 - Designed and executed advanced text preprocessing pipelines including tokenization, stopword removal, dictionary-based lemmatization, and TF-IDF feature extraction to optimize vocabulary size.
 - Trained, evaluated, and fine-tuned Support Vector Machine (SVM), Logistic Regression, and Naïve Bayes classifiers, mitigating severe class imbalance through engineered class-weight balancing and n-gram analysis.
 - Conducted exploratory data analysis (EDA) by plotting category-specific word clouds and profiling class distributions to visualize distinguishing lexical features and quantify severe dataset imbalances.
 - Optimized hyperparameter configurations to achieve peak classification accuracy and F1 scores, directly validating performance for down-stream content recommendation engines. Achieved a grade of 100%.